Design Requirements Document (DRD)

Project: REB AMI — Advanced Metering Infrastructure

Version: 1.0 | Date: April 2026

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1. Document Control

1.1 Revision History

Version	Date	Author	Changes
0.1	March 2026	OTBL	Initial draft
1.0	April 2026	OTBL	Approved release

1.2 References

Document	Version	Description
REB-RFP-2026-001	1.0	Original RFP
OTBL-PROP-2026-AMI	2.0	Technical proposal
DEV-INTIMATION-001	1.0	Development intimation

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2. System Overview

2.1 Purpose

This DRD defines the functional and non-functional requirements, system architecture, data models, and interface specifications for the REB AMI system.

2.2 System Context

The REB AMI system serves as the central platform for:

- Collecting meter data from smart meters across REB service areas

- Processing and validating meter readings
- Generating billing data and synchronizing with Oracle ERP
- Providing real-time monitoring and analytics dashboards
- Enabling mobile access for field operations

2.3 System Boundaries

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┌──────────────────────────────────────────────────────────┐
│                      REB AMI SYSTEM                      │
│  ┌──────────────┐  ┌──────────────┐  ┌──────────────────┐ │
│  │ Web Portal   │  │ Mobile App   │  │ Background Services │ │
│  └──────────────┘  └──────────────┘  └──────────────────┘ │
│  ┌──────────────────────────────────────────────────┐   │
│  │              Core API Services                   |   |
│  └──────────────────────────────────────────────────┘   │
│  ┌──────────────────────────────────────────────────┐   │
│  │         PostgreSQL Database + Cache Layer        |   |
│  └──────────────────────────────────────────────────┘   │
└──────────────────────────────────────────────────────────┘
        |                                  |
        ▼                                  ▼
┌──────────────────┐                  ┌──────────────────────┐
|  Smart Meters    |                  |  Oracle ERP System   |
|  (External)      |                  |  (External)          |
└──────────────────┘                  └──────────────────────┘
```

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3. Functional Requirements

3.1 User Management Module (FR-UM)

Req ID	Requirement	Priority	Acceptance Criteria
FR-UM-001	System shall support role-based access control	High	Min 5 roles configurable
FR-UM-002	Users shall authenticate via username/password	High	Secure login within 3 sec
FR-UM-003	System shall support password reset via email	Medium	Reset link valid 24 hours
FR-UM-004	Admin shall manage user accounts (CRUD)	High	Full lifecycle management
FR-UM-005	System shall log all user activities	High	Audit trail with timestamps

Req ID	Requirement	Priority	Acceptance Criteria
FR-UM-006	Support two-factor authentication	Medium	OTP via SMS/Email

3.2 Meter Data Collection Module (FR-MDC)

Req ID	Requirement	Priority	Acceptance Criteria
FR-MDC-001	Collect meter readings via DLMS/COSEM protocol	High	Support IEC 62056 standard
FR-MDC-002	Store readings with meter ID, timestamp, values	High	No data loss guaranteed
FR-MDC-003	Validate readings against threshold rules	High	Flag anomalies automatically
FR-MDC-004	Support manual reading entry for offline meters	Medium	Field entry via mobile app
FR-MDC-005	Handle data backfill for communication gaps	High	Automated gap detection
FR-MDC-006	Support interval data (15-min, hourly, daily)	High	Configurable intervals

3.3 Billing Integration Module (FR-BI)

Req ID	Requirement	Priority	Acceptance Criteria
FR-BI-001	Calculate consumption from meter readings	High	kWh accuracy ±0.1%
FR-BI-002	Apply tariff structures to consumption	High	Support tiered rates
FR-BI-003	Generate billing determinants for Oracle	High	Daily batch export
FR-BI-004	Sync customer master data from Oracle	High	Real-time + batch sync
FR-BI-005	Handle billing exceptions and adjustments	Medium	Approval workflow
FR-BI-006	Support prepaid meter balance management	Medium	Real-time balance updates

3.4 Dashboard & Reporting Module (FR-DR)

Req ID	Requirement	Priority	Acceptance Criteria
FR-DR-001	Display real-time system health metrics	High	Dashboard refresh < 5 sec
FR-DR-002	Show consumption trends and analytics	High	Interactive charts
FR-DR-003	Generate scheduled reports (PDF/Excel)	Medium	Email delivery support
FR-DR-004	Provide outage detection and alerts	High	Alert within 5 minutes
FR-DR-005	Support custom report builder	Low	Drag-and-drop interface
FR-DR-006	Geographic visualization (GIS maps)	Medium	Cluster + heatmap views

3.5 Mobile Application Module (FR-MA)

Req ID	Requirement	Priority	Acceptance Criteria
FR-MA-001	Support Android 8.0+ and iOS 13+	High	Native-like performance
FR-MA-002	Enable offline data collection	High	Sync when online
FR-MA-003	Capture meter photos with GPS tagging	Medium	Image compression
FR-MA-004	Display work orders for field staff	High	Push notifications
FR-MA-005	Consumer self-service (view bills, usage)	Medium	Secure consumer login
FR-MA-006	Biometric authentication support	Low	Fingerprint/Face ID

4. Non-Functional Requirements

4.1 Performance Requirements (NFR-P)

Req ID	Requirement	Target	Measurement
NFR-P-001	API response time	< 200ms (95th percentile)	Application monitoring
NFR-P-002	Dashboard load time	< 3 seconds	Browser performance
NFR-P-003	Concurrent users	500 simultaneous	Load testing

Req ID	Requirement	Target	Measurement
NFR-P-004	Data processing rate	10,000 readings/minute	Throughput testing
NFR-P-005	Database query time	< 100ms (complex queries)	Query profiling

4.2 Scalability Requirements (NFR-S)

Req ID	Requirement	Target
NFR-S-001	Support horizontal scaling	Add app servers without downtime
NFR-S-002	Handle 1 million meters	Within 5-year growth plan
NFR-S-003	Database partitioning	Time-based partitioning for readings
NFR-S-004	Microservices-ready	Modular service architecture

4.3 Security Requirements (NFR-SEC)

Req ID	Requirement	Implementation
NFR-SEC-001	Data encryption at rest	AES-256 encryption
NFR-SEC-002	Data encryption in transit	TLS 1.3
NFR-SEC-003	API authentication	JWT tokens with refresh
NFR-SEC-004	SQL injection prevention	Parameterized queries
NFR-SEC-005	OWASP Top 10 compliance	Security testing
NFR-SEC-006	Audit logging	Immutable audit trail
NFR-SEC-007	Data backup	Daily backup, 30-day retention

4.4 Availability Requirements (NFR-A)

Req ID	Requirement	Target
NFR-A-001	System uptime	99.5% (excluding planned maintenance)
NFR-A-002	Planned maintenance window	Sunday 2-6 AM

Req ID	Requirement	Target
NFR-A-003	Recovery Time Objective (RTO)	4 hours
NFR-A-004	Recovery Point Objective (RPO)	1 hour

5. System Architecture Design

5.1 Architecture Principles

- **Layered Architecture**: Separation of presentation, business logic, and data
- **Stateless Services**: Enable horizontal scaling
- **API-First Design**: RESTful APIs for all integrations
- **Event-Driven**: Asynchronous processing for heavy workloads
- **Security by Design**: Defense in depth approach

5.2 Component Architecture

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┌──────────────────────────────────────────────────────────────────────┐
│                          PRESENTATION LAYER                          │
│  ┌──────────────────┐  ┌──────────────────┐                         │
│  │   Vue.js Web     │  │  React Native    │                         │
│  │   Dashboard      │  │  Mobile App      │                         │
│  └──────────┬───────┘  └──────────┬───────┘                         │
└─────────────┼─────────────────────┼─────────────────────────────────┘
              │                     │
              ▼                     ▼
┌──────────────────────────────────────────────────────────────────────┐
│                          API GATEWAY (NGINX)                         │
│  • SSL Termination  • Rate Limiting  • Load Balancing                │
└──────────────────────────────┬───────────────────────────────────────┘
                               │
┌──────────────────────────────┼───────────────────────────────────────┐
│                          APPLICATION LAYER                           │
│  ┌──────────────┐  ┌──────────────┐  ┌──────────────────────┐       │
│  │ Auth         │  │ Meter Data   │  │ Billing              │       │
│  │ Service      │  │ Service      │  │ Service              │       │
│  └──────────────┘  └──────────────┘  └──────────────────────┘       │
│  ┌──────────────┐  ┌──────────────┐  ┌──────────────────────┐       │
│  │ Reporting    │  │ Notification│  │ Integration          │       │
```

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│  │ Service     │  │ Service     │  │ Service              │    │
│  └─────────────┘  └─────────────┘  └──────────────────────┘    │
└────────────────────────────┬────────────────────────────────────┘
                             │
┌────────────────────────────┴────────────────────────────────────┐
│                        DATA LAYER                               │
│  ┌──────────────────┐  ┌──────────────────┐                    │
│  │   PostgreSQL     │  │   Redis Cache    │                    │
│  │   (Primary DB)   │  │   (Session/Data) │                    │
│  └──────────────────┘  └──────────────────┘                    │
└────────────────────────────────────────────────────────────────┘
```

5.3 Service Definitions

Service	Responsibility	Technology
Auth Service	Authentication, authorization, user management	Node.js + JWT
Meter Data Service	Reading collection, validation, storage	Node.js + PostgreSQL
Billing Service	Consumption calculation, tariff application	Node.js + PostgreSQL
Reporting Service	Analytics, report generation	Node.js + Chart.js
Notification Service	Email, SMS, push notifications	Node.js + Firebase
Integration Service	Oracle sync, external API management	Node.js + Oracle Client

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6. Data Model Design

6.1 Entity Relationship Overview

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┌──────────────┐      ┌──────────────┐      ┌──────────────┐
│   USERS      │      │  CUSTOMERS   │      │   METERS     │
├──────────────┤      ├──────────────┤      ├──────────────┤
│ user_id (PK)│      │ cust_id (PK)│──────▶│ meter_id(PK)│
│ email        │      │ name         │      │ cust_id(FK)  │
│ role_id(FK)  │      │ address      │      │ meter_type   │
│ status       │      │ phone        │      │ install_date│
└──────────────┘      └──────────────┘      └──────┬───────┘
                                                    │
```

```
                    ┌──────────────────────────┘
                    ▼

┌──────────────────────────────────┐      ┌──────────────────────┐
│       METER_READINGS             │      │    BILLING_RECORDS   │
├──────────────────────────────────┤      ├──────────────────────┤
│ reading_id (PK)                  │      │ bill_id (PK)         │
│ meter_id (FK)                    │──────▶│ meter_id (FK)        │
│ reading_timestamp                │      │ billing_period       │
│ active_energy_kwh                │      │ consumption_kwh      │
│ reactive_energy_kvarh            │      │ amount_due           │
│ voltage, current, power_factor   │      │ status               │
│ validation_status                │      │ oracle_sync_status   │
└──────────────────────────────────┘      └──────────────────────┘
```

6.2 Key Tables Specification

6.2.1 meters

Column	Type	Constraints	Description
meter_id	UUID	PK	Unique meter identifier
customer_id	UUID	FK	Reference to customer
serial_number	VARCHAR(50)	UNIQUE, NOT NULL	Manufacturer serial
meter_type	ENUM	NOT NULL	SMART, PREPAID, CONVENTIONAL
communication_type	ENUM		GPRS, RF, PLC
installation_date	DATE		Meter installation date
location_lat	DECIMAL(10,8)		GPS latitude
location_lng	DECIMAL(11,8)		GPS longitude
status	ENUM	NOT NULL	ACTIVE, INACTIVE, FAULTY
created_at	TIMESTAMP	DEFAULT NOW()	Record creation
updated_at	TIMESTAMP		Last modification

6.2.2 meter_readings

Column	Type	Constraints	Description
reading_id	UUID	PK	Unique reading identifier
meter_id	UUID	FK, INDEX	Reference to meter
reading_timestamp	TIMESTAMP	NOT NULL, INDEX	Reading capture time
active_energy_import	DECIMAL(12,3)		kWh imported
active_energy_export	DECIMAL(12,3)		kWh exported (solar)
reactive_energy	DECIMAL(12,3)		kVARh
voltage_l1	DECIMAL(6,2)		Phase 1 voltage
voltage_l2	DECIMAL(6,2)		Phase 2 voltage
voltage_l3	DECIMAL(6,2)		Phase 3 voltage
current_l1	DECIMAL(8,3)		Phase 1 current
power_factor	DECIMAL(4,3)		Power factor
validation_status	ENUM		VALID, SUSPECT, ESTIMATED
source	ENUM		AUTO, MANUAL, ESTIMATED

Table partitioned by reading_timestamp (monthly)

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7. Interface Specifications

7.1 External Interfaces

7.1.1 Oracle Integration Interface

Attribute	Specification
Protocol	Oracle DB Link + REST API
Authentication	Service account with encrypted credentials
Data Format	JSON (API), Oracle native (DB Link)
Sync Frequency	Real-time (critical) + Daily batch

Sync Operations:

Operation	Direction	Frequency	Data
Customer Master	Oracle → AMI	Hourly	Customer details
Billing Determinants	AMI → Oracle	Daily	Consumption data
Payment Updates	Oracle → AMI	Real-time	Payment status
Tariff Structure	Oracle → AMI	On change	Rate schedules

7.1.2 Smart Meter Interface

Attribute	Specification
Protocol	DLMS/COSEM over GPRS/TCP
Standard	IEC 62056-21, IEC 62056-46
Security	DLMS Security Suite 1 (AES-GCM-128)
Data Objects	Active/Reactive Energy, Voltage, Current, Events

7.2 API Specifications

7.2.1 RESTful API Standards

- Base URL: `https://api.reb-ami.gov.bd/v1`
- Authentication: Bearer JWT token
- Content-Type: application/json

- Versioning: URL path versioning

7.2.2 Core API Endpoints

Endpoint	Method	Description
`/auth/login`	POST	User authentication
`/auth/refresh`	POST	Token refresh
`/meters`	GET	List meters (paginated)
`/meters/{id}`	GET	Get meter details
`/meters/{id}/readings`	GET	Get meter readings
`/readings`	POST	Submit meter reading
`/billing/calculate`	POST	Calculate consumption
`/reports/generate`	POST	Generate report
`/dashboard/metrics`	GET	Dashboard KPIs

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8. Security Design

8.1 Authentication Flow

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┌─────────┐      ┌──────────┐      ┌──────────────┐      ┌──────────┐
│ Client │──────▶│  NGINX  │──────▶│ Auth        │──────▶│ User DB  │
└─────────┘      └──────────┘      │ Service     │      └──────────┘
    │              │              └──────┬──────┘
    │              │                     │
    │              │              ┌──────▼──────┐
    │              │              │ JWT Token   │
    │◀─────────────┼──────────────│ Generated   │
    │              │              └──────────────┘
    │              │
    │   [Subsequent Requests with JWT]
    │──────────────▶
```

8.2 Authorization Matrix

Role	Meters	Readings	Billing	Reports	Users	System
Super Admin	CRUD	CRUD	CRUD	CRUD	CRUD	CRUD
Admin	CRUD	CRUD	CRU	CRUD	CRU	R
Operator	RU	CR	R	R	-	-
Field Staff	R	CR	-	-	-	-
Consumer	R (own)	R (own)	R (own)	-	-	-

C=Create, R=Read, U=Update, D=Delete

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9. Deployment Architecture

9.1 Environment Strategy

Environment	Purpose	Infrastructure
Development	Feature development	Local Docker
Staging	Integration testing	Single VM
UAT	User acceptance	Prod-like (scaled down)
Production	Live system	Full HA cluster

9.2 Production Deployment

```
┌──────────────────┐
│   CloudFlare     │
│   (CDN + WAF)    │
└──────────┬───────┘
           │
           ▼
┌──────────────────┐
│     NGINX        │
│  Load Balancer   │
└──────┬───────────┘
```

```
                  |         |
          ┌───────▼─┐   ┌─▼───────┐
          │ App VM1 │   │ App VM2 │
          │ Node.js │   │ Node.js │
          └─────┬───┘   └───┬─────┘
                |           |
              ┌─▼───────────▼─┐
              | PostgreSQL    |
              | (Primary)     |
              └───────┬───────┘
                      |
              ┌───────▼───────┐
              | PostgreSQL    |
              | (Replica)     |
              └───────────────┘
```

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10. Acceptance Criteria

10.1 Functional Acceptance

- [ ] All FR requirements implemented and tested
- [ ] User roles and permissions working correctly
- [ ] Meter data collection operational
- [ ] Oracle integration verified
- [ ] Mobile app published to stores

10.2 Non-Functional Acceptance

- [ ] Performance benchmarks met (NFR-P)
- [ ] Security audit passed
- [ ] 99.5% uptime during pilot period
- [ ] Documentation complete

10.3 Sign-off Requirements

Stakeholder	Criteria	Status
REB IT	Technical compliance	Pending
REB Operations	Functional compliance	Pending

Stakeholder	Criteria	Status
OTBL QA	Test completion	Pending
Security Team	Security audit	Pending

11. Appendices

Appendix A: Glossary

Term	Definition
AMI	Advanced Metering Infrastructure
DLMS	Device Language Message Specification
COSEM	Companion Specification for Energy Metering
MDM	Meter Data Management
HES	Head-End System

Appendix B: Document Approval

Role	Name	Signature	Date
REB Project Sponsor			
OTBL Technical Lead			
OTBL Project Manager			

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